Large Language Models: Prompting

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Contents

Introduction

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Introduction

연세대학교에 대해 알려줘

연세대학교(Yonsei University)는 1885년에 설립된 대한민국의 대표적인 사립 종합대학교로, 서울특별시 서대문구 신촌동에 본교를 두고 있습니다. 서울대학교, 고려대학교와 함께 'SKY'로 불리며 국내에서 가장 명망 있는 대학 중 하나로 평가받고 있습니다. 위키백과

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Introduction

- **Language models can solve computer tasks (Kim et al., 2023)**

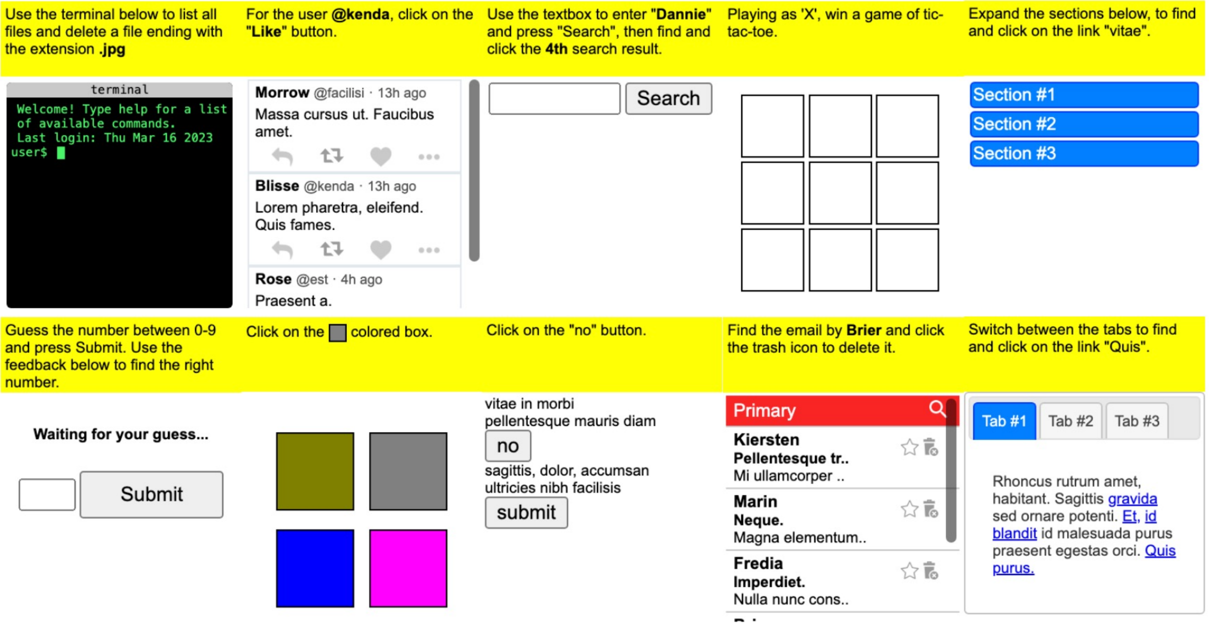


Figure 1: MiniWoB++ environment. Every task contains a natural language prompt in yellow. The agent then uses keyboard strokes and mouse clicks to accomplish the task.

Introduction

- **LLMs can also automate web navigation (Zhou et al., 2023)**

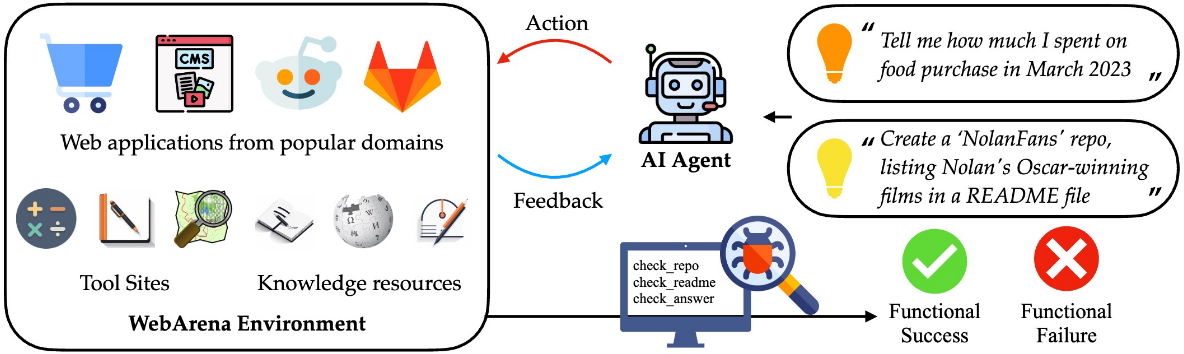
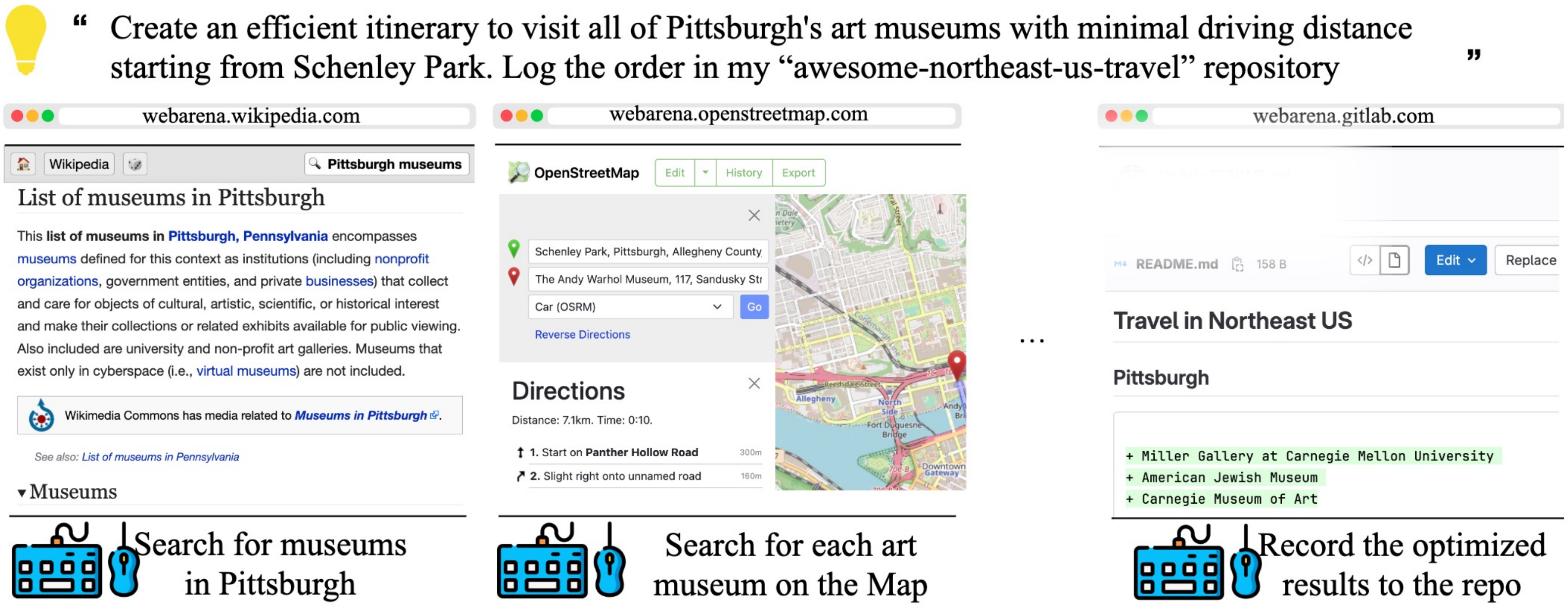


Figure 1: `WebArena` is a standalone, self-hostable web environment for building autonomous agents. `WebArena` creates websites from four popular categories with functionality and data mimicking their real-world equivalents. To emulate human problem-solving, `WebArena` also embeds tools and knowledge resources as independent websites. `WebArena` introduces a benchmark on interpreting *high-level realistic* natural language command to concrete web-based interactions. We provide validators to programmatically validate the functional correctness of each task.

Introduction

- **LLMs can also automate web navigation (Zhou et al., 2023)**



Introduction

- **LLMs can also be clinical reasoners (Kwon et al., 2023)**

Large Language Models Are Clinical Reasoners: Reasoning-Aware Diagnosis Framework with Prompt-Generated Rationales

Taeyoon Kwon[1*], Kai Tzu-iunn Ong[1*], Dongjin Kang[2], Seungjun Moon[2], Jeong Ryong Lee[3], Dosik Hwang[3], Beomseok Sohn[4], Yongsik Sim[4], Dongha Lee[1], Jinyoung Yeo[1]

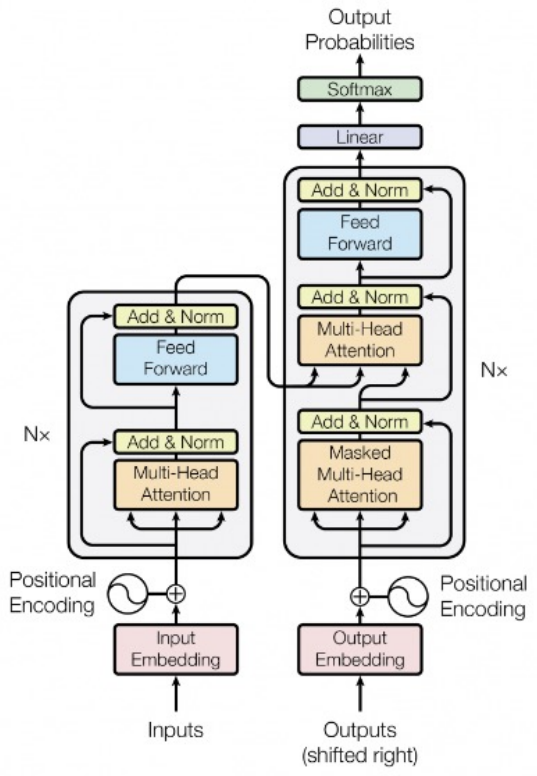
[1]Department of Artificial Intelligence, Yonsei University
[2]Department of Computer Science and Engineering, Yonsei University
[3]Department of Electrical and Electronic Engineering, Yonsei University
[4]Department of Radiology, College of Medicine, Yonsei University

Introduction

- **Let’s see how this progress become possible within just a few years!!!**

Recap

Recap – Transformer



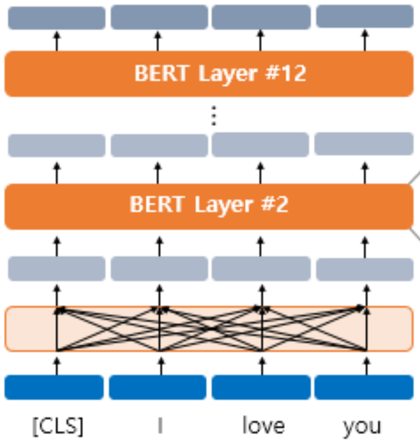
- **Attention is All you Need (2017)**
- **Why Transformer?**
  - Self-Attention
  - No recurrence
  - Scalability
  - Parallelizability

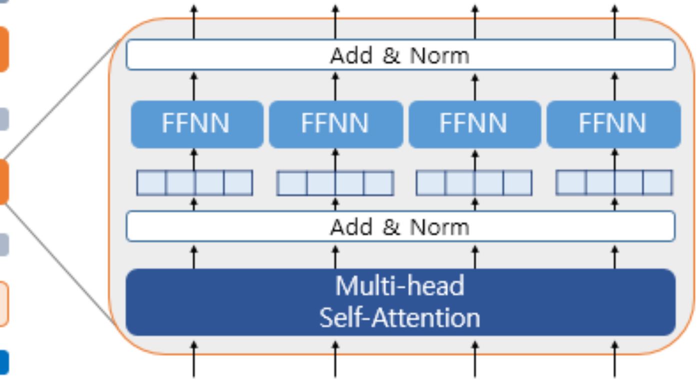
  ...

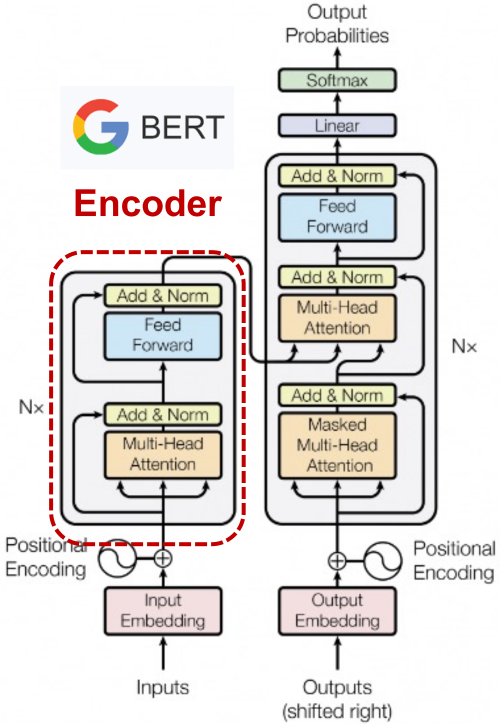
➔ **The foundation of nearly all modern DL models**

Vaswani et al. "Attention is All you Need", NeurIPS 2017

Recap – BERT



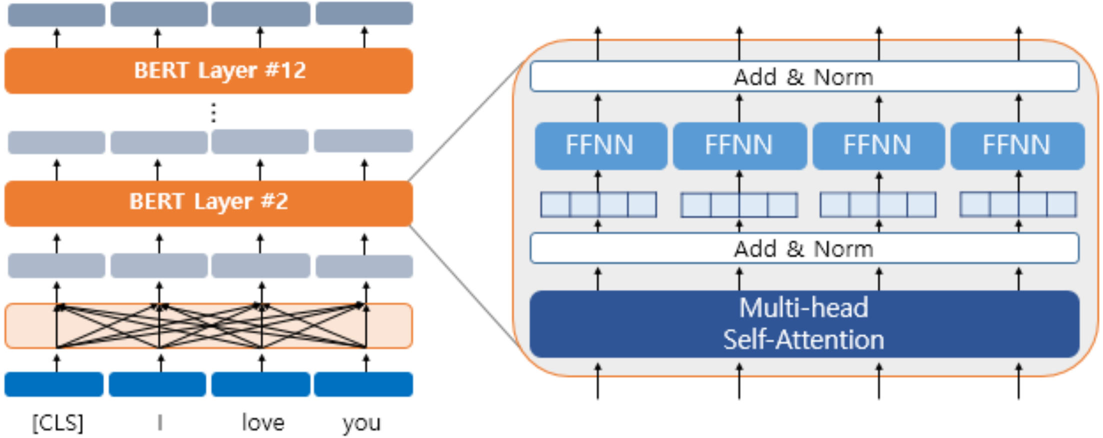




Devlin, Jacob et al. "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding", NAACL 2019

Recap – BERT

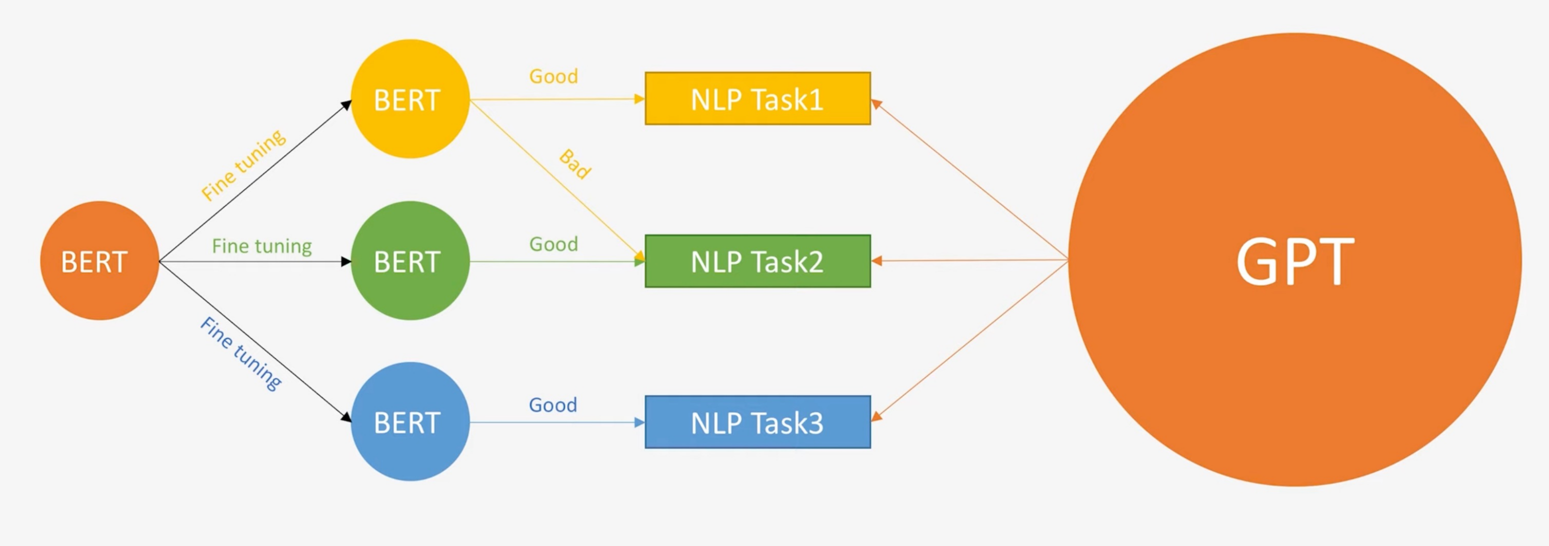
- **BERT is a bidirectional encoder model**
  - Trained with **two objectives**:
    - **Masked Language Modeling (MLM)**
    - **Next Sentence Prediction (NSP)**
  - Excels at natural language understanding tasks with full sentence context
  - Fine-tuned for each downstream task (e.g., QA, classification, NER)



Devlin, Jacob et al. "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding", NAACL 2019

Recap – GPT

- **GPT is a general-purpose model**
  - Trained with **single objective**: **next-token prediction**
  - Performs well on many tasks with **zero-shot** or **few-shot prompting**



Radford et al. "Language models are unsupervised multitask learners", 2019

Recap – GPT

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- **Input = Task + Context**
  - Output → predict the most probable next token
- **GPT-2's improved performance enables it to handle multiple tasks without fine-tuning — using just prompts.**

$$p(output|input) \longrightarrow p(output|input, task)$$

"Brevet Sans Garantie Du Gouvernement", translated to English: → GPT → "Patented without government warranty".

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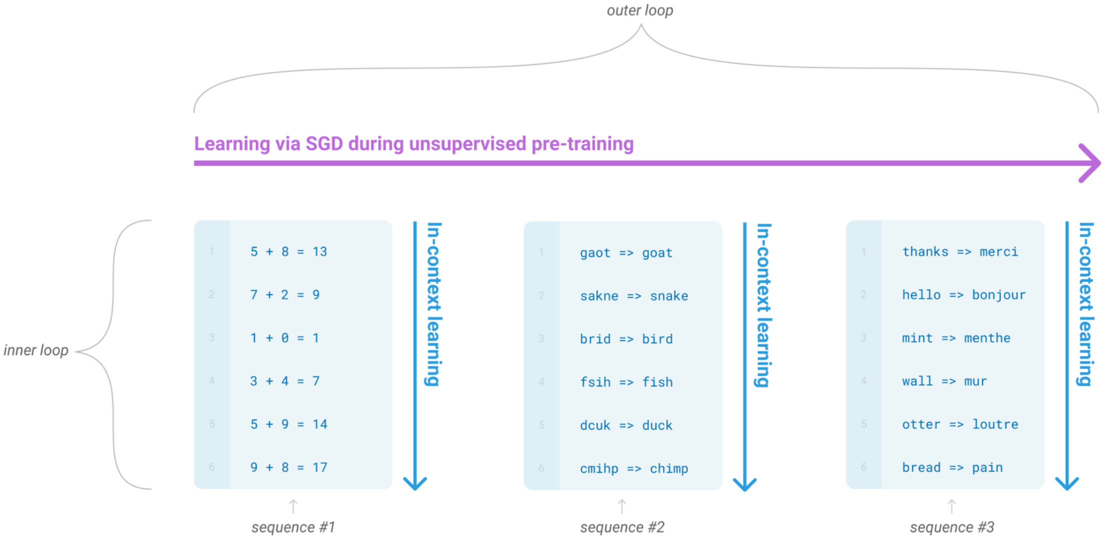
Emergent of Large Language Models

In-Context Learning

In-Context Learning

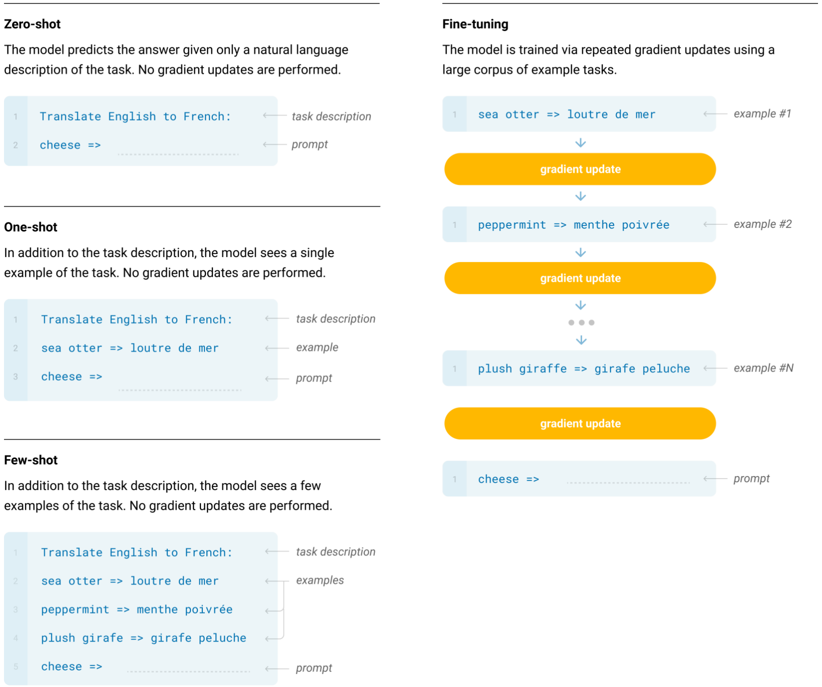
- **Task Learning via Context, not Parameters**
  - Language models can learn from few-shot demonstrations (**Without parameter update**)
  - Previous: Fine-tuning (**Difficult and expensive** to collect training datasets)
  - Emergent phenomenon in LLMs



Brown et al. "Language Models are Few-Shot Learners", NeurIPS 2020

In-Context Learning

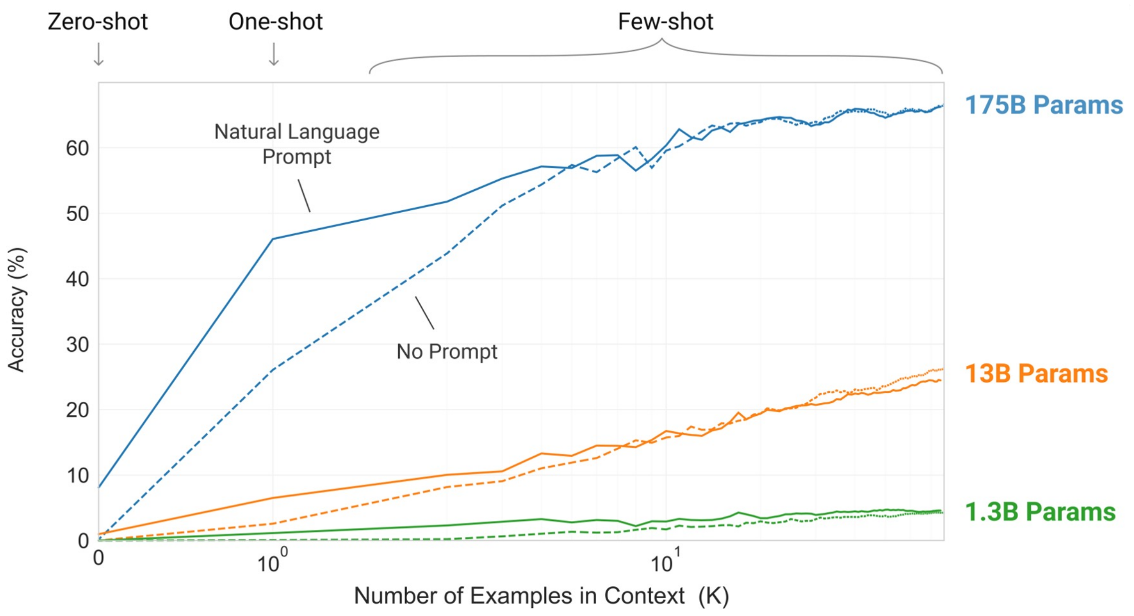
- **Learning "how to learn"**



Brown et al. "Language Models are Few-Shot Learners", NeurIPS 2020

In-Context Learning

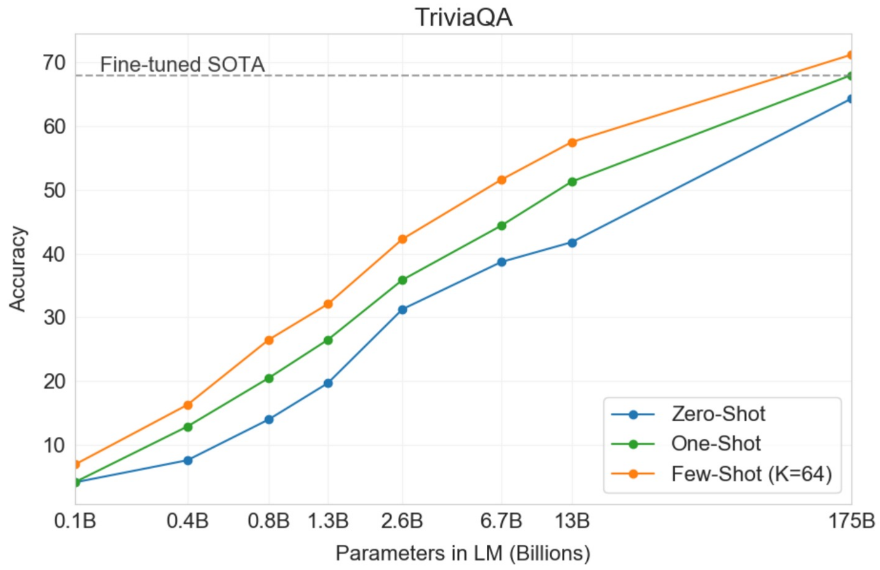
- **The larger the model is, the better the accuracy is**



Brown et al. "Language Models are Few-Shot Learners", NeurIPS 2020

In-Context Learning

- **Few-shot > One-shot > Zero-shot**



Brown et al. “Language Models are Few-Shot Learners”, NeurIPS 2020

Task 1-1: In-Context Learning

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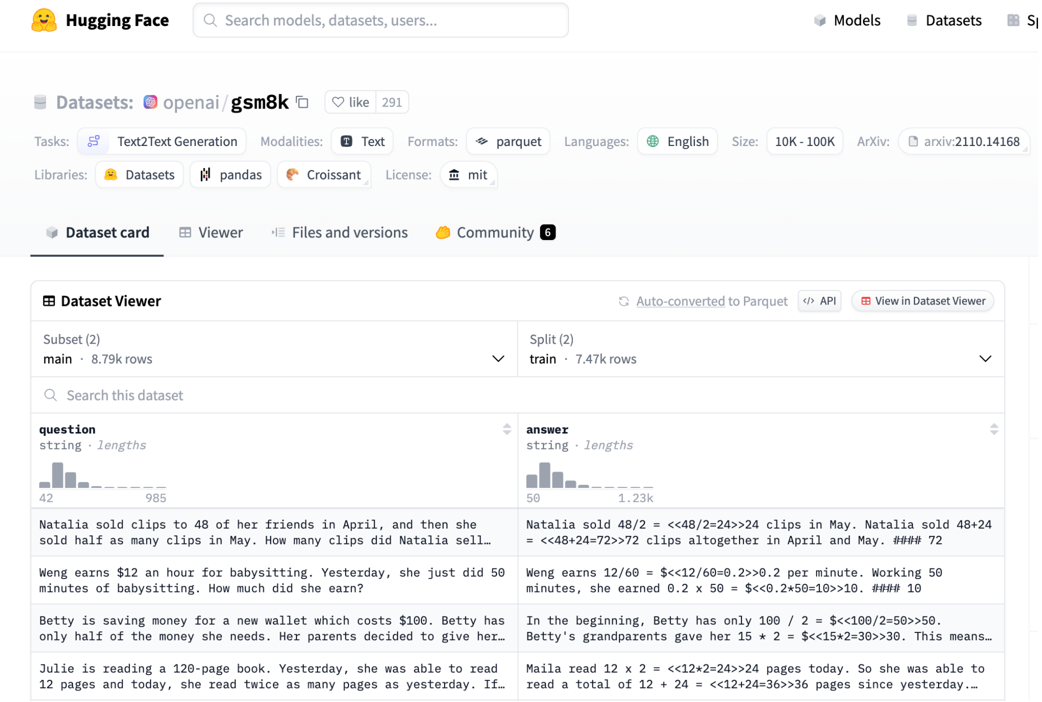
- **Task: Mathematical Reasoning (GSM8K)**



https://huggingface.co/datasets/openai/gsm8k

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Task 1-1: In-Context Learning

- **Examples**

Question: Natalia sold clips to 48 of her friends in April, and then she sold half as many clips in May. How many clips did Natalia sell altogether in April and May?

Answer: Natalia sold 48/2 = <<48/2=24>>24 clips in May. Natalia sold 48+24 = <<48+24=72>>72 clips altogether in April and May. #### 72

https://huggingface.co/datasets/openai/gsm8k

Task 1-1: In-Context Learning

- **In-context demonstration construction**

[Example 1]
Input:
~~~
Output:
~~~
…

[Example 4]
Input:
~~~
Output:

https://huggingface.co/datasets/openai/gsm8k
~~~

Task 1-1: In-Context Learning

- **In-context demonstration construction**

[Example 1]
Input:
~~~
Output:
~~~
…

[Example 4]
Input:
~~~
Output: ⇨ **LLM will generate output after this identifier!**

https://huggingface.co/datasets/openai/gsm8k
~~~

Grok API

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- **Through Grok API, you can get generation results of various chat models.**

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Instruction Tuning

Challenges of Pre-trained LLMs

- **Even LLMs suffer to understand human's intention and value**
  - LLM's are trained to predict next token
  - Low-quality data in web

Prompt *Explain the moon landing to a 6 year old in a few sentences.*

Completion GPT-3

```
Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.
```

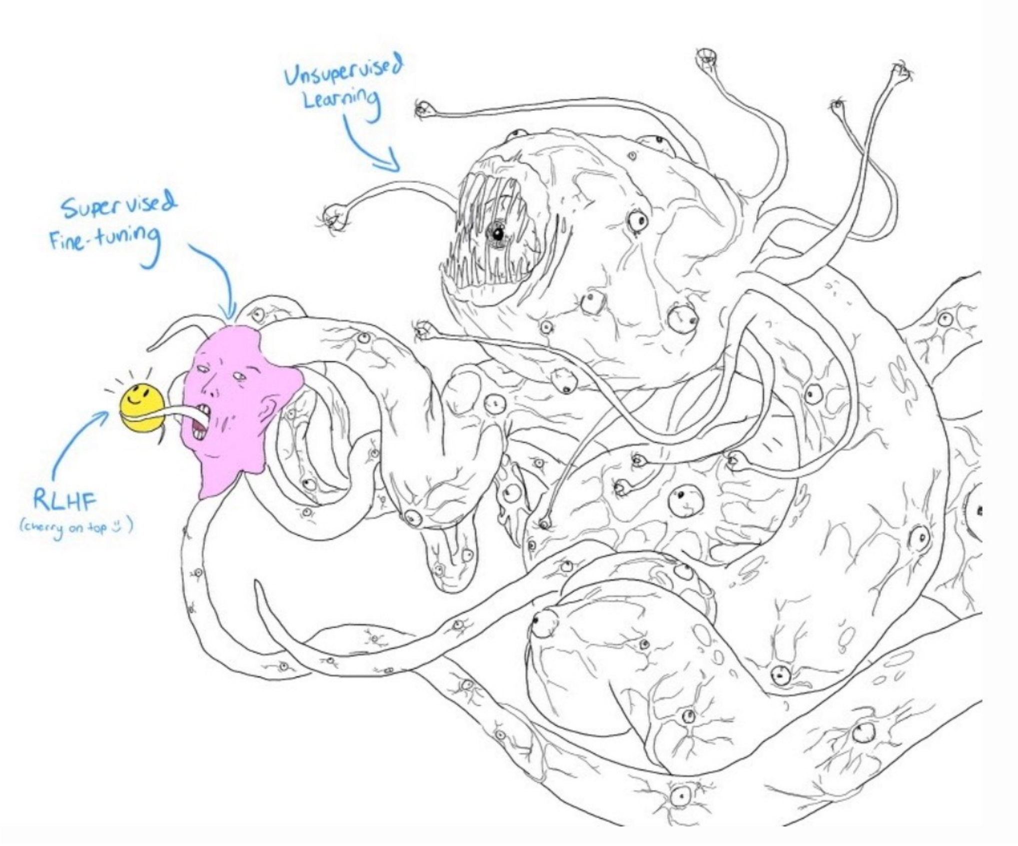
InstructGPT

```
People went to the moon, and they took pictures of what they saw, and sent them
back to the earth so we could all see them.
```

Ouyang et al., "Training Language Models to Follow Instructions with Human Feedback", NeurIPS 2022

https://openai.com/index/instruction-following/

Challenges of Pre-trained LLMs



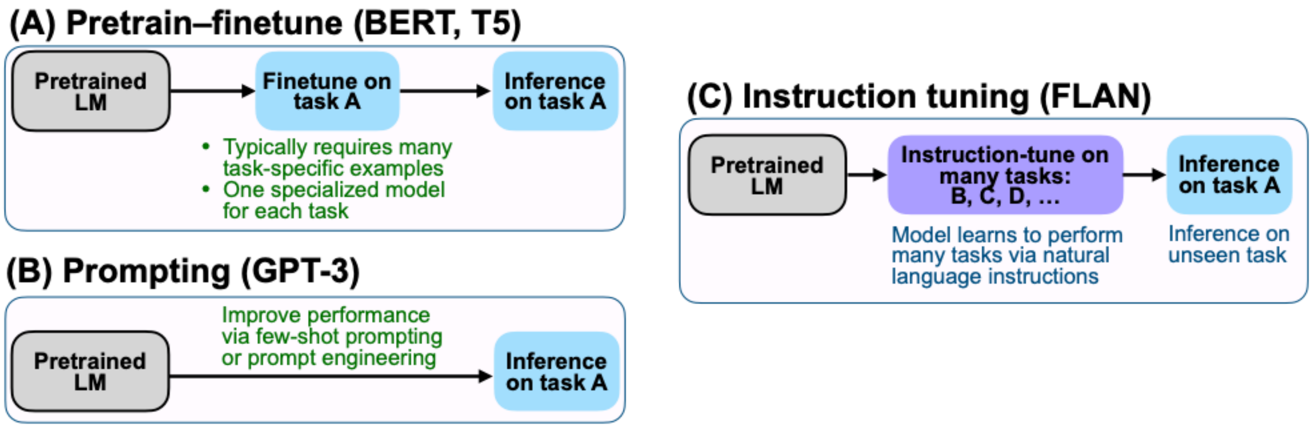
https://openai.com/index/instruction-following/

Instruction Tuning

- **Previous: Learning mapping from input to output without instruction**

Instruction Tuning

- Previous: Learning mapping from input to output without instruction
- **Now: Training LMs to follow instructions that describe the tasks asked for LMs to complete**



Instruction Tuning

- **Instruction following capabilities are generalized to unseen tasks**

Resource →	SUP-NATINST (this work)	NATINST (Mishra et al., 2022b)	CROSSFIT (Ye et al., 2021)	PROMPTSOURCE (Bach et al., 2022)	FLAN (Wei et al., 2022)	INSTRUCTGPT (Ouyang et al., 2022)
Has task instructions?	✓	✓	✗	✓	✓	✓
Has negative examples?	✓	✓	✗	✗	✗	✗
Has non-English tasks?	✓	✗	✗	✗	✓	✓
Is public?	✓	✓	✓	✓	✓	✗
Number of tasks	1616	61	269	176	62	–
Number of instructions	1616	61	–	2052	620	14378
Number of annotated tasks types	76	6	13	13*	12	10
Avg. task definition length (words)	56.6	134.4	–	24.8	8.2	–

(a) SUP-NATINST (this work) (b) NATINST (c) PROMPTSOURCE (T0 subset) (d) FLAN (e) INSTRUCTGPT

Task 1-2: Instruction following LMs

- **Task: Add instruction and also see the zero-shot performance!**

Instruction: ~~~

[Example 1]
Input:
~~~
Output:
~~~
…

[Example 4]
Input:
~~~
Output:
~~~

Task 1-2: Instruction following LMs

- **Task: Add instruction and also see the zero-shot performance!**

Instruction: ~~~

[Example 1]
Input:
~~~
Output:

Zero shot!
~~~

How can we better use LLMs?

Chain-of-Thought Prompting

Chain-of-Thought Prompting

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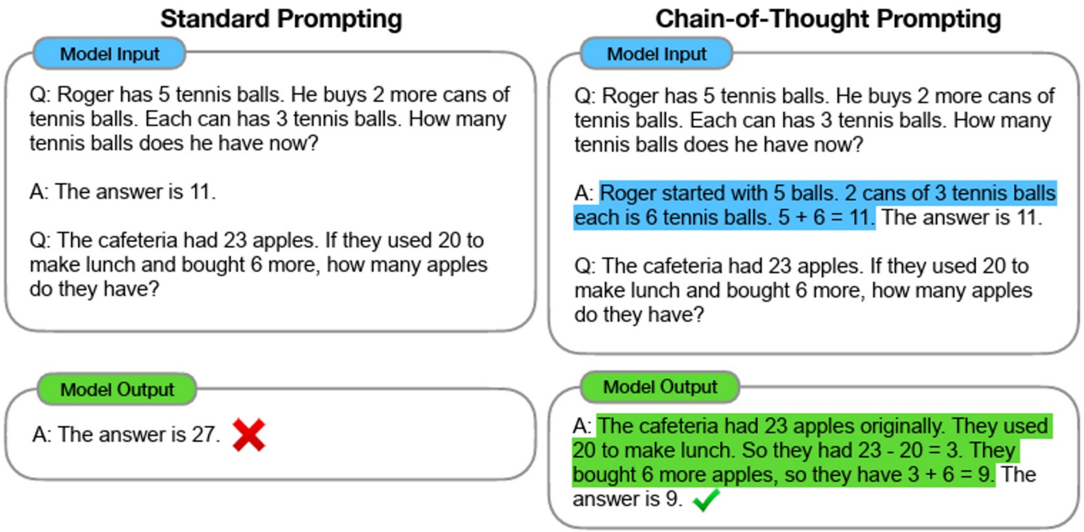
- **LLMs can improve their reasoning capabilities through intermediate natural language reasoning steps before generating answer**
  - Emergent phenomenon in LLMs



Wei et al. "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models", NeurIPS 2022

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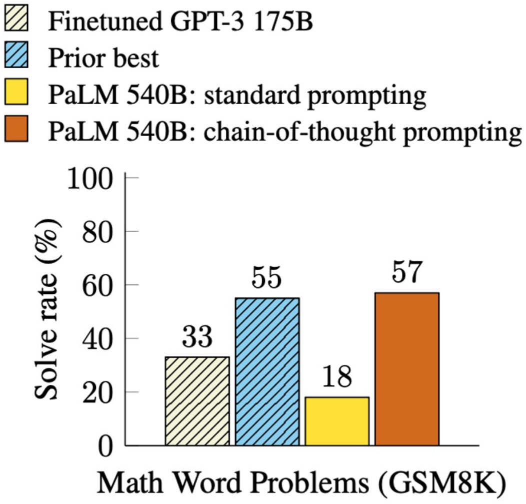
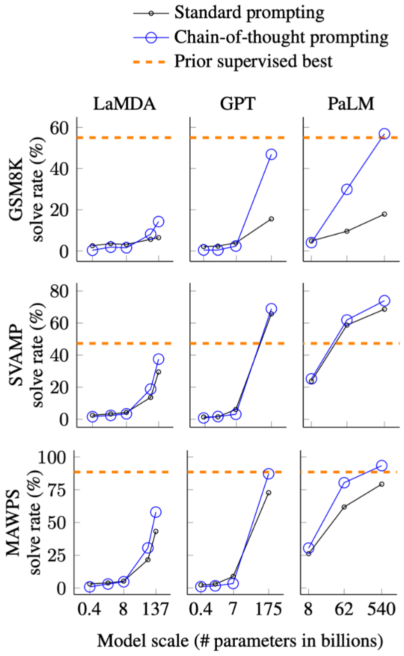
Chain-of-Thought Prompting

- **CoT outperforms standard prompting by a large margin and achieves new state-of-the-art performance.**
  - Even better performance than fine-tuned LLMs.



Chain-of-Thought Prompting

- **CoT is an emergent phenomenon of the LLMs**
  - Model scale



Wei et al. “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models”, NeurIPS 2022

Chain-of-Thought Prompting

- **How to generate intermediate reasoning steps?**
  - Few shot demonstrations!!!

PROMPT FOR AQUA ALGEBRAIC WORD PROBLEMS

Q: John found that the average of 15 numbers is 40. If 10 is added to each number then the mean of the numbers is?
Answer Choices: (a) 50 (b) 45 (c) 65 (d) 78 (e) 64

A: If 10 is added to each number, then the mean of the numbers also increases by 10. So the new mean would be 50. The answer is (a).

Q: If a / b = 3/4 and 8a + 5b = 22,then find the value of a.
Answer Choices: (a) 1/2 (b) 3/2 (c) 5/2 (d) 4/2 (e) 7/2

A: If a / b = 3/4, then b = 4a / 3. So 8a + 5(4a / 3) = 22. This simplifies to 8a + 20a / 3 = 22, which means 44a / 3 = 22. So a is equal to 3/2. The answer is (b).

Q: A person is traveling at 20 km/hr and reached his destiny in 2.5 hr then find the distance?
Answer Choices: (a) 53 km (b) 55 km (c) 52 km (d) 60 km (e) 50 km

A: The distance that the person traveled would have been 20 km/hr * 2.5 hrs = 50 km. The answer is (e).

Q: How many keystrokes are needed to type the numbers from 1 to 500?
Answer Choices: (a) 1156 (b) 1392 (c) 1480 (d) 1562 (e) 1788

A: There are 9 one-digit numbers from 1 to 9. There are 90 two-digit numbers from 10 to 99. There are 401 three-digit numbers from 100 to 500. 9 + 90(2) + 401(3) = 1392. The answer is (b).

Wei et al. "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models", NeurIPS 2022

Task 2-1: CoT Prompting

- **Solve GSM8K with Chain-of-Thought!!**
  - Better with instruction??

[Example 1]
Input:
~~~
Output:
In the beginning, Betty has only 100 / 2 = $&lt;&lt;100/2=50&gt;&gt;50.
Betty's grandparents gave her 15 * 2 = $&lt;&lt;15*2=30&gt;&gt;30.
This means, Betty needs 100 - 50 - 30 - 15 = $&lt;&lt;100-50-30-15=5&gt;&gt;5 more.
#### 5
…

[Example 4]
Input:
~~~
Output:

Task 2-1: CoT Prompting

- **How about Zero-shot CoT??**
  - Add the instruction?

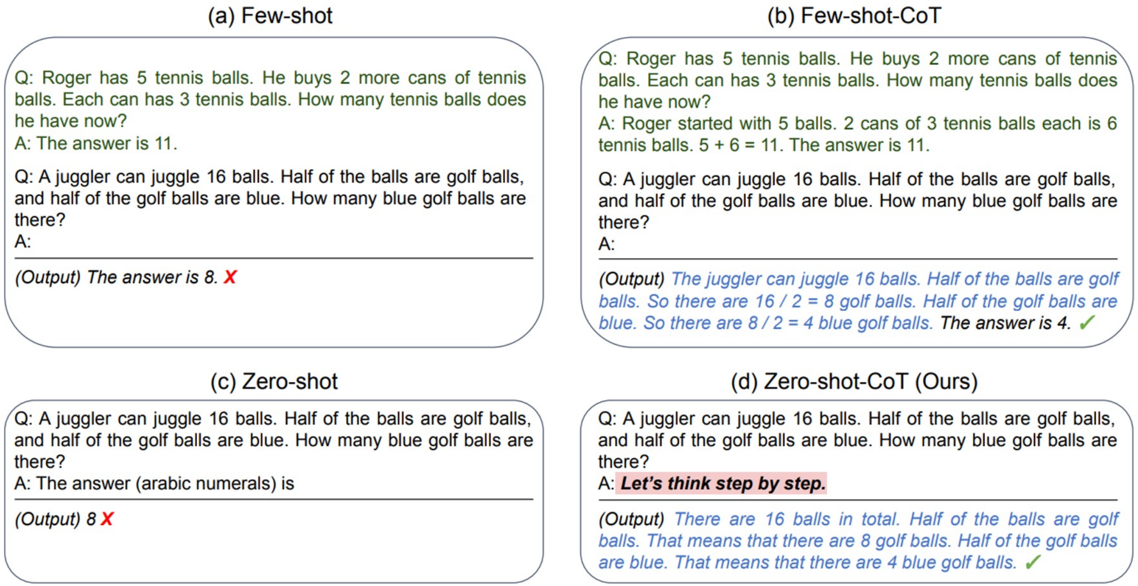
Instruction: Generate the answer with rationale~~~

[Example 1]
Input:
~~~
Output:
~~~

Zero-shot
Chain-of-Thought Prompting

Zero-shot CoT

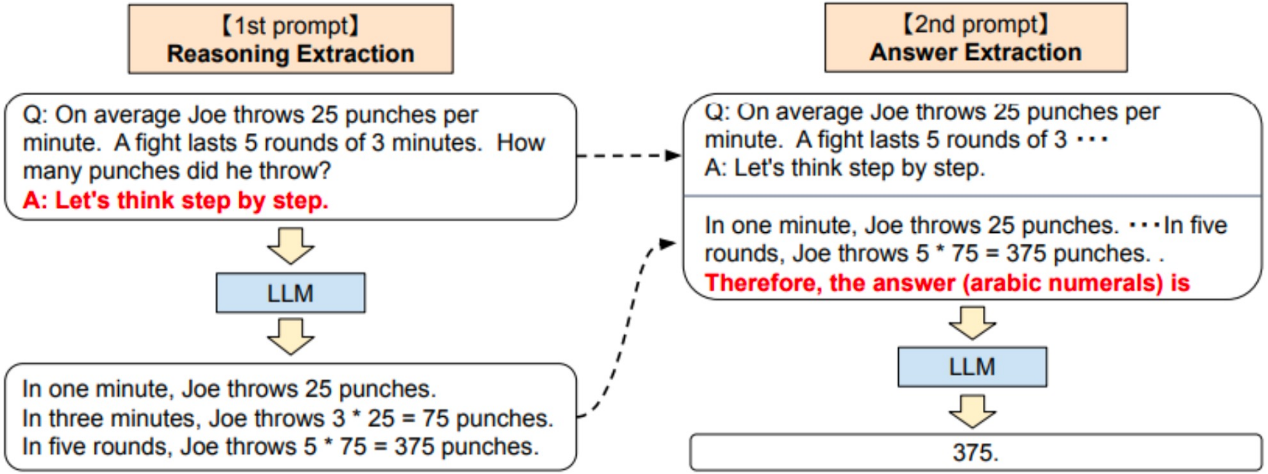
- **How can we generate reasoning steps without any demonstrations?**



Kojima et al. "Large language models are zero-shot reasoners.", NeurIPS 2022

Zero-shot CoT

- **Two stage prompting**
  - Reasoning extraction and answer extraction



Kojima et al. "Large language models are zero-shot reasoners.", NeurIPS 2022

Zero-shot CoT

- **Two stage prompting**
  - Reasoning extraction and answer extraction

	MultiArith	GSM8K
Zero-Shot	**17.7**	**10.4**
Few-Shot (2 samples)	33.7	15.6
Few-Shot (8 samples)	33.8	15.6
Zero-Shot-CoT	**78.7**	**40.7**
Few-Shot-CoT (2 samples)	84.8	41.3
Few-Shot-CoT (4 samples : First) (*1)	89.2	-
Few-Shot-CoT (4 samples : Second) (*1)	90.5	-
Few-Shot-CoT (8 samples)	93.0	48.7
Zero-Plus-Few-Shot-CoT (8 samples) (*2)	**92.8**	**51.5**
Finetuned GPT-3 175B [Wei et al., 2022]	-	33
Finetuned GPT-3 175B + verifier [Wei et al., 2022]	-	55
PaLM 540B: Zero-Shot	**25.5**	**12.5**
PaLM 540B: Zero-Shot-CoT	**66.1**	**43.0**
PaLM 540B: Zero-Shot-CoT + self consistency	**89.0**	**70.1**
PaLM 540B: Few-Shot [Wei et al., 2022]	-	17.9
PaLM 540B: Few-Shot-CoT [Wei et al., 2022]	-	56.9
PaLM 540B: Few-Shot-CoT + self consistency [Wang et al., 2022]	-	74.4

Task 2-1: CoT Prompting

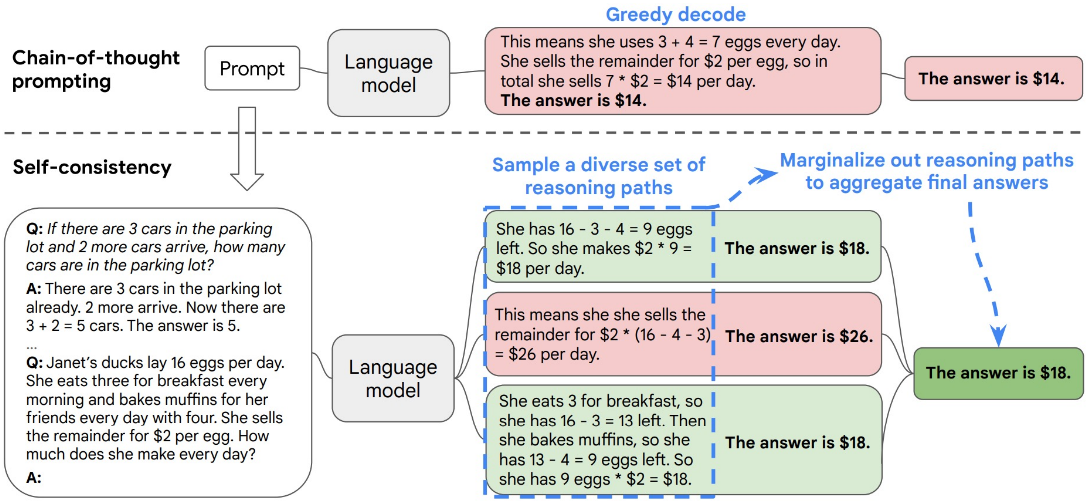
- **Zero-shot CoT with reasoning extraction!**

[Example 1]
Input:
~~~
Output:

[Answer]
**Let's think step by step.**
~~~

Self-Consistency

Self-Consistency

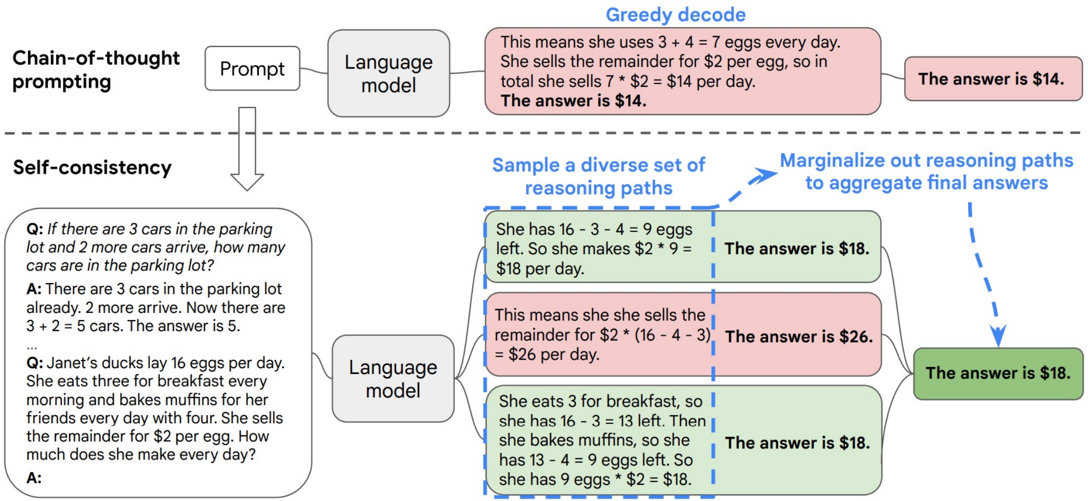
- **CoT decoding strategy: Greedy decoding**



Wang et al. "Self-Consistency Improves Chain of Thought Reasoning in Language Models.", ICLR 2023

Self-Consistency

- **Complex reasoning tasks admit multiple reasoning paths**
  - The more deliberate thinking is required, the greater the diversity of reasoning path is.



Wang et al. "Self-Consistency Improves Chain of Thought Reasoning in Language Models.", ICLR 2023

Self-Consistency

- **Self-consistency significantly improves the reasoning performance**
  - Mathematical reasoning

	Method	AddSub	MultiArith	ASDiv	AQuA	SVAMP	GSM8K
	Previous SoTA	**94.9**[a]	60.5[a]	75.3[b]	37.9[c]	57.4[d]	35[e] / 55[g]
UL2-20B	CoT-prompting	18.2	10.7	16.9	23.6	12.6	4.1
	Self-consistency	24.8 (+6.6)	15.0 (+4.3)	21.5 (+4.6)	26.9 (+3.3)	19.4 (+6.8)	7.3 (+3.2)
LaMDA-137B	CoT-prompting	52.9	51.8	49.0	17.7	38.9	17.1
	Self-consistency	63.5 (+10.6)	75.7 (+23.9)	58.2 (+9.2)	26.8 (+9.1)	53.3 (+14.4)	27.7 (+10.6)
PaLM-540B	CoT-prompting	91.9	94.7	74.0	35.8	79.0	56.5
	Self-consistency	93.7 (+1.8)	99.3 (+4.6)	81.9 (+7.9)	48.3 (+12.5)	86.6 (+7.6)	74.4 (+17.9)
GPT-3 Code-davinci-001	CoT-prompting	57.2	59.5	52.7	18.9	39.8	14.6
	Self-consistency	67.8 (+10.6)	82.7 (+23.2)	61.9 (+9.2)	25.6 (+6.7)	54.5 (+14.7)	23.4 (+8.8)
GPT-3 Code-davinci-002	CoT-prompting	89.4	96.2	80.1	39.8	75.8	60.1
	Self-consistency	91.6 (+2.2)	**100.0** (+3.8)	**87.8** (+7.6)	**52.0** (+12.2)	**86.8** (+11.0)	**78.0** (+17.9)

Self-Consistency

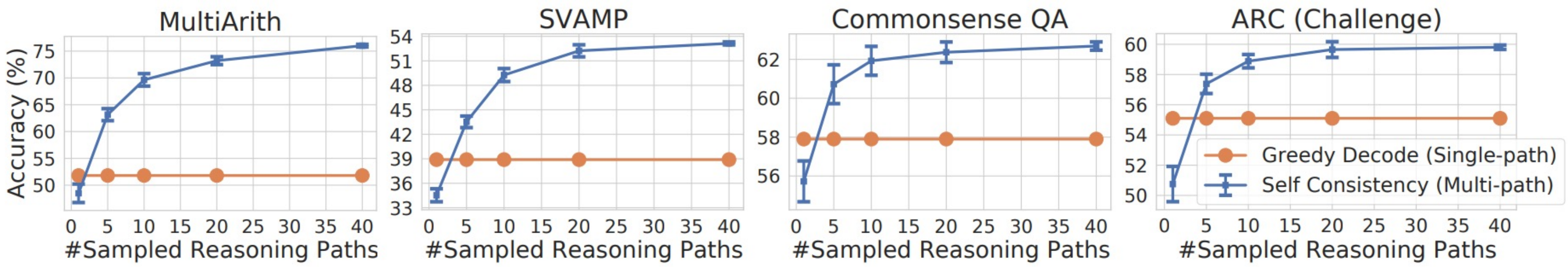
- **Self-consistency significantly improves the reasoning performance**
  - Commonsense reasoning

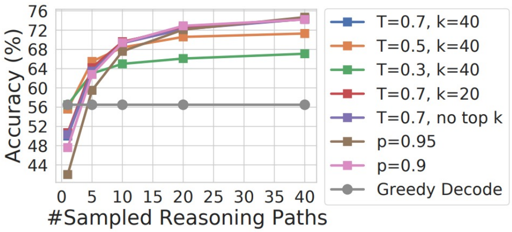
	Method	CSQA	StrategyQA	ARC-e	ARC-c	Letter (4)	Coinflip (4)
	Previous SoTA	**91.2**[a]	73.9[b]	86.4[c]	75.0[c]	N/A	N/A
UL2-20B	CoT-prompting	51.4	53.3	61.6	42.9	0.0	50.4
	Self-consistency	55.7 (+4.3)	54.9 (+1.6)	69.8 (+8.2)	49.5 (+6.8)	0.0 (+0.0)	50.5 (+0.1)
LaMDA-137B	CoT-prompting	57.9	65.4	75.3	55.1	8.2	72.4
	Self-consistency	63.1 (+5.2)	67.8 (+2.4)	79.3 (+4.0)	59.8 (+4.7)	8.2 (+0.0)	73.5 (+1.1)
PaLM-540B	CoT-prompting	79.0	75.3	95.3	85.2	65.8	88.2
	Self-consistency	80.7 (+1.7)	**81.6** (+6.3)	**96.4** (+1.1)	**88.7** (+3.5)	70.8 (+5.0)	91.2 (+3.0)
GPT-3 Code-davinci-001	CoT-prompting	46.6	56.7	63.1	43.1	7.8	71.4
	Self-consistency	54.9 (+8.3)	61.7 (+5.0)	72.1 (+9.0)	53.7 (+10.6)	10.0 (+2.2)	75.9 (+4.5)
GPT-3 Code-davinci-002	CoT-prompting	79.0	73.4	94.0	83.6	70.4	99.0
	Self-consistency	81.5 (+2.5)	79.8 (+6.4)	96.0 (+2.0)	87.5 (+3.9)	**73.4** (+3.0)	**99.5** (+0.5)

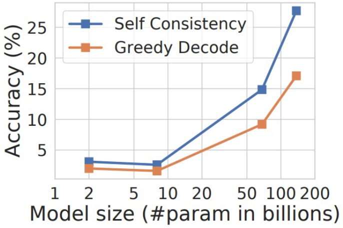
Wang et al. "Self-Consistency Improves Chain of Thought Reasoning in Language Models.", ICLR 2023

Self-Consistency

- **Self-consistency significantly improves the reasoning performance**
  - Robust to the number of reasoning paths, sampling strategy and scaling







Wang et al. "Self-Consistency Improves Chain of Thought Reasoning in Language Models.", ICLR 2023

Task 2-2: Self-consistency

- **Let's apply self-consistency!**
  - How? Sample the examples multiple times!

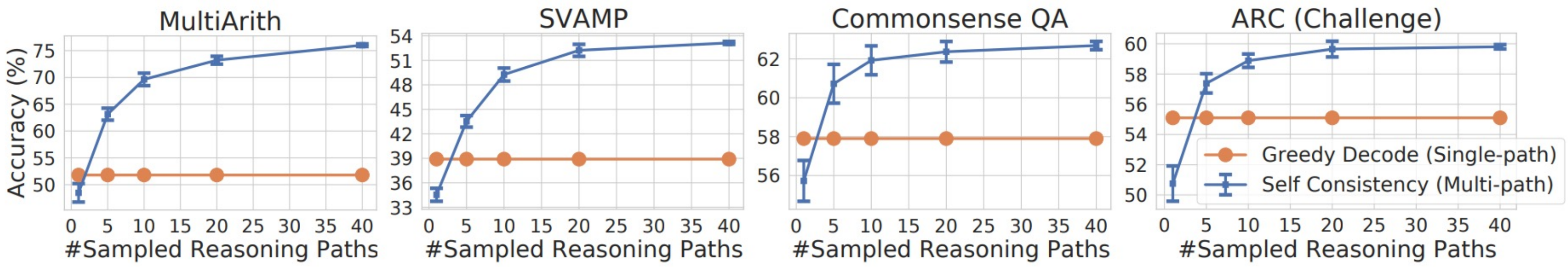
[Example 1]
Input:
~~~
Output:
~~~
…

[Example 4]
Input:
~~~
Output:

[Example 1*]
Input:
~~~
Output:
~~~
…

[Example 4*]
Input:
~~~
Output:

…

Task 2-2: Self-consistency

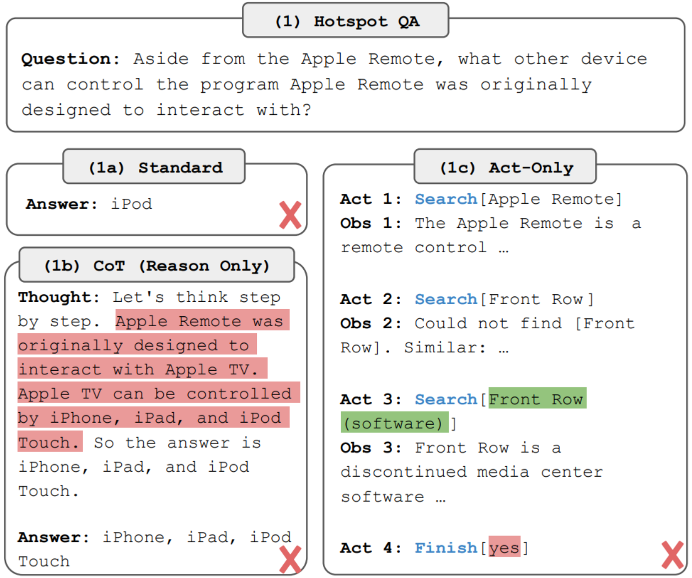
- **Let's plot the result**
  - 1~5 times



ReAct

ReAct

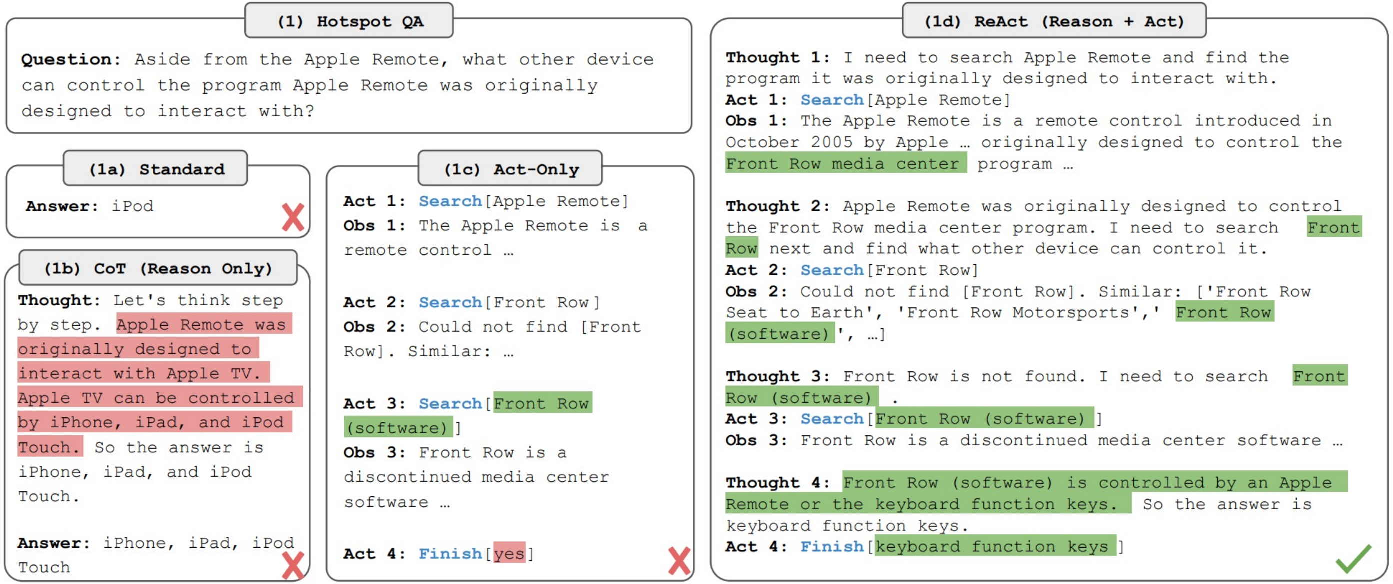
- **Recent approaches**
  - **Reasoning only**: Internal representation only, not grounded in external world
    -> hallucination, error propagation
  - **Actions**: generate domain-specific actions or plans, and then use a controller to choose or execute them -> No reasoning abstractly about high-level goals



Yao et al. "ReAct: Synergizing Reasoning and Acting in Language Models", ICLR 2023

ReAct

- **Rea**soning + **Act**ing
  - Prompt LLMs to generate both verbal reasoning traces and actions pertaining to a task



ReAct

(1) Hotspot QA

Question: Aside from the Apple Remote, what other device can control the program Apple Remote was originally designed to interact with?

(1a) Standard

Answer: iPod ✗

(1b) CoT (Reason Only)

Thought: Let's think step by step. Apple Remote was originally designed to interact with Apple TV. Apple TV can be controlled by iPhone, iPad, and iPod Touch. So the answer is iPhone, iPad, and iPod Touch.

Answer: iPhone, iPad, iPod Touch ✗

(1c) Act-Only

Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control …

Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: …

Act 3: Search[Front Row (software)]
Obs 3: Front Row is a discontinued media center software …

Act 4: Finish[yes] ✗

(1d) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.
Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple … originally designed to control the Front Row media center program …

Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search Front Row next and find what other device can control it.
Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports',' Front Row (software) ', …]

Thought 3: Front Row is not found. I need to search Front Row (software) .
Act 3: Search[Front Row (software)]
Obs 3: Front Row is a discontinued media center software …

Thought 4: Front Row (software) is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.
Act 4: Finish[keyboard function keys] ✓

Yao et al. "ReAct: Synergizing Reasoning and Acting in Language Models", ICLR 2023

ReAct

- **Results**
  - HotpotQA: CoT outperforms
  - Fever: Fact verification -> ReAct outperforms
  - SC: Self-consistency

Prompt Method[a]	HotpotQA (EM)	Fever (Acc)
Standard	28.7	57.1
CoT (Wei et al., 2022)	29.4	56.3
CoT-SC (Wang et al., 2022a)	33.4	60.4
Act	25.7	58.9
ReAct	27.4	60.9
CoT-SC → ReAct	34.2	**64.6**
ReAct → CoT-SC	**35.1**	62.0
Supervised SoTA[b]	67.5	89.5

ReAct

- **ReAct significantly mitigates the hallucination issue of CoT**

	Type	Definition	ReAct	CoT
Success	True positive	Correct reasoning trace and facts	94%	86%
	False positive	Hallucinated reasoning trace or facts	6%	14%
Failure	Reasoning error	Wrong reasoning trace (including failing to recover from repetitive steps)	47%	16%
	Search result error	Search return empty or does not contain useful information	23%	-
	Hallucination	Hallucinated reasoning trace or facts	0%	56%
	Label ambiguity	Right prediction but did not match the label precisely	29%	28%

Yao et al. "ReAct: Synergizing Reasoning and Acting in Language Models", ICLR 2023

Task 2-3: ReAct

- **Task: Wikipedia-based question answering task**
  - HotpotQA

Datasets: hotpotqa/**hotpot_qa** like 127 Follow hotpotqa 12

Tasks: Question Answering Languages: English Size: 100K<n<1M ArXiv: arxiv:1809.09600 Tags: multi-hop

Dataset card | Files and versions | Community 7

Dataset Viewer

The viewer is disabled because this dataset repo requires arbitrary Python code execution. Please consider removing the loading script and relying on automated data support (you can use convert to parquet from the datasets library). If this is not possible, please open a discussion for direct help.

Dataset Card for "hotpot_qa"

Dataset Summary

HotpotQA is a new dataset with 113k Wikipedia-based question-answer pairs with four key features: (1) the questions require finding and reasoning over multiple supporting documents to answer; (2) the questions are diverse and not constrained to any pre-existing knowledge bases or knowledge schemas; (3) we provide sentence-level supporting facts required for reasoning, allowingQA systems to reason with strong supervision and explain the predictions; (4) we offer a new type of factoid comparison questions to test QA systems' ability to extract relevant facts and perform necessary comparison.

Task 2-3: ReAct

- **What to do**
  - Let's compare the performance of CoT / Act / ReAct